

Let X be a real-valued random variable.

Definition 1 (CDF and PDF). The **cumulative distribution function (CDF)** of X is defined as $F_X(x) := \Pr\{X \leq x\}$ for all $x \in \mathbb{R}$. A **probability density function (PDF)** of X is a nonnegative measurable function f_X satisfying $F_X(x) = \int_{-\infty}^x f_X(t) dt$ for all $x \in \mathbb{R}$. Such a density exists if and only if F_X is absolutely continuous. In that case, $f_X(x) = F'_X(x)$ for almost every x .

Remark 1. The distribution of X is characterized by its **CDF**, so the **CDF** exists for any random variable, whereas the **PDF** may not exist for all random variables. For example, the **CDF** of a discrete random variable has jumps and cannot be represented as an integral of a density.

Definition 2 (Moment). The k -th moment of a random variable X is defined as $\mathbb{E}(X^k)$, provided that the expectation exists.

Remark 2. The mean is the first moment, and the variance is the second *central* moment, defined as $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2]$. The second moment of X is $\mathbb{E}X^2$.

Definition 3 (Sample). A sample is a collection of random variables, often denoted as $\{X_i\}_{i=1}^n$ where n is the sample size.

Definition 4 (Statistic). A statistic is a function of a sample.

Definition 5 (Estimator). An estimator is a statistic used to guess, compute, or estimate a parameter.

Remark 3. An *estimate* is the realized value of an estimator given a realized sample. An *estimand* is the parameter that an estimator aims to estimate.

Definition 6 (Parameter). A parameter is a target feature of the underlying model.

Remark 4 (What is a parameter really?). There are different ways to define a parameter. In class, we defined a parameter as a function of the population distribution, which is a definition commonly used in statistics. Under this definition, identification is built in: once the population distribution is given, the parameter is uniquely determined. In econometrics, we also work with *economic* models, whose parameters may describe preferences, technologies, or behavioral responses. Different values of these parameters can sometimes generate the same distribution of observable variables, so identification becomes a separate question.

Definition 7 (independent and identically distributed). A sample $\{X_i\}_{i=1}^n$ is said to be **independent and identically distributed (iid)** if the X_i are independent and identically distributed, denoted as $\{X_i\}_{i=1}^n \stackrel{\text{iid}}{\sim} F$ for some distribution F .

Remark 5. Here is an example of a sample that is identically distributed but *not* independent: Let $\{Z_i\}_{i=0}^n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, with $n \geq 2$. Define $X_i = (Z_0 + Z_i)/\sqrt{2}$ for all $i = 1, \dots, n$. Then each $X_i \sim \mathcal{N}(0, 1)$, but $\text{Cov}(X_i, X_j) = 1/2$ for $i \neq j$ because they share the term Z_0 . Thus, the X_i are identically distributed but not independent.

Definition 8 (Identification). A parameter is said to be identified if it can be written as a function of the population distribution, i.e., if it can be uniquely determined from the population distribution of the observable variables.

Remark 6 (Alternative Formulation). You will often see identification formulated in the following way: Let Θ denote the parameter space and let P_θ denote the population distribution of the observable variables under parameter $\theta \in \Theta$. Then the parameter θ is said to be identified if $P_\theta = P_{\theta'}$ implies $\theta = \theta'$ for all $\theta, \theta' \in \Theta$.

Example 1 (Schooling and Wages). Suppose that log wages Y are determined by years of schooling S and an unobserved factor U :

$$Y = \alpha + \beta S + U, \quad \mathbb{E}(U) = 0. \quad (1)$$

Here β describes the effect of an additional year of schooling holding U fixed. We observe (Y, S) , but not U , which may include unobserved ability. Assume finite first moments and place no further restrictions on the relationship between S and U . For any candidate value b , define

$$\begin{aligned} a_b &= \mathbb{E}(Y) - b \mathbb{E}(S), \\ U_b &= Y - a_b - bS. \end{aligned}$$

Then $Y = a_b + bS + U_b$ and $\mathbb{E}(U_b) = 0$. Thus, every value b is compatible with exactly the same joint distribution of the observables (Y, S) . The schooling effect β is not identified under these assumptions: Even knowing the entire population distribution would not distinguish it from the contribution of the unobserved factor. This demonstrates that only requiring $\mathbb{E}(U) = 0$ is not enough to identify β , since an arbitrary relationship between S and U is allowed.

Acronyms

CDF cumulative distribution function. [1](#)

iid independent and identically distributed. [2](#)

PDF probability density function. [1](#)